

OPTIMIZATION UTILITIES USER DOCUMENT

Three utilities have been developed to execute and structure output from SUSTAIN's two-tier optimization: (1) the Inclusive Best Solutions Utility, (2) the MJR BMP Utility, and (3) the Two-tiered Optimization Utility. These utilities accept and process output from SUSTAIN model runs such that their results can be uploaded to the Optimization Viewer. The Optimization Viewer visualizes the full range of BMP configurations (as defined in the BMP Opportunities Utility) and their corresponding cost-effectiveness to discern optimal BMP performance.

Figure 1 illustrates the relative position of the three optimization utilities in the WMMS 2.0 Optimization Framework. Reference the *WMMS 2.0 Phase II Report: BMP Model and Optimization Framework* for additional details on the model framework and its full suite of utilities. As indicated in the figure, input to the optimization utilities is the output from the BMP Opportunities Utility, which generates SUSTAIN input files from user-provided BMP specifications (e.g., drainage area, maximum footprints).

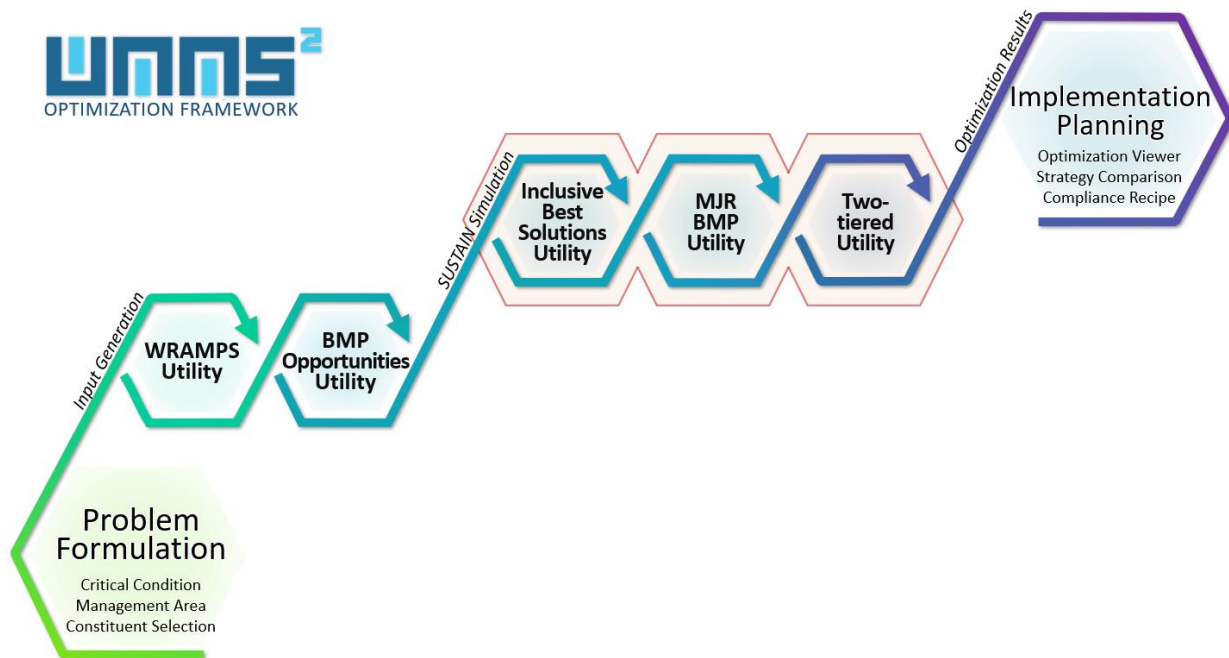


Figure 1. Flow diagram of the WMMS 2.0 Optimization Framework with the three optimization utilities in red.

To support the use of the optimization utilities for model simulation, this document describes:

- ▼ The two-tier optimization approach
- ▼ Inputs, processing, and output for the Tier 1 optimization
- ▼ Inputs, processing, and output for the Tier 2 optimization
- ▼ Inputs, processing, and output for the MJR BMP Utility
- ▼ Inputs, processing, and output for the Two-tiered Optimization Utility
- ▼ Utility Troubleshooting

1 TWO-TIER OPTIMIZATION APPROACH

The SUSTAIN model is used for process-based continuous BMP simulation at an hourly (or finer) timestep. In the first tier of the optimization, SUSTAIN's integrated cost-benefit BMP module uses optimization algorithms to search through a user-defined range of BMP opportunities and evaluate their associated cost and benefits (e.g., volume reduction). This process results in a “cloud” of possible BMP opportunities for implementation, each of which can be directly compared for differences in performance. Figure 1-1 provides an example of 20,000 BMP scenarios considered for an individual subwatershed/jurisdiction (“jurished”). These scenarios are based on user-defined management objectives (e.g., volume capture for the 85th percentile storm) as well as BMP attributes (e.g., drainage area, maximum footprint). With these modeling bounds, the BMP module directly evaluates screened management opportunities in each jurished. Each of the small gray circles (i.e., “All Solutions” points) in Figure 1-1 represent a potential management plan that comprises a unique combination of BMPs within the jurished. The larger “Best Solutions” points are the most cost-effective solutions within the jurished—the curve represents the highest achievable performance at every cost interval. For every simulated jurished in a management area, a cloud of solutions and an accompanying best solutions curve is generated.

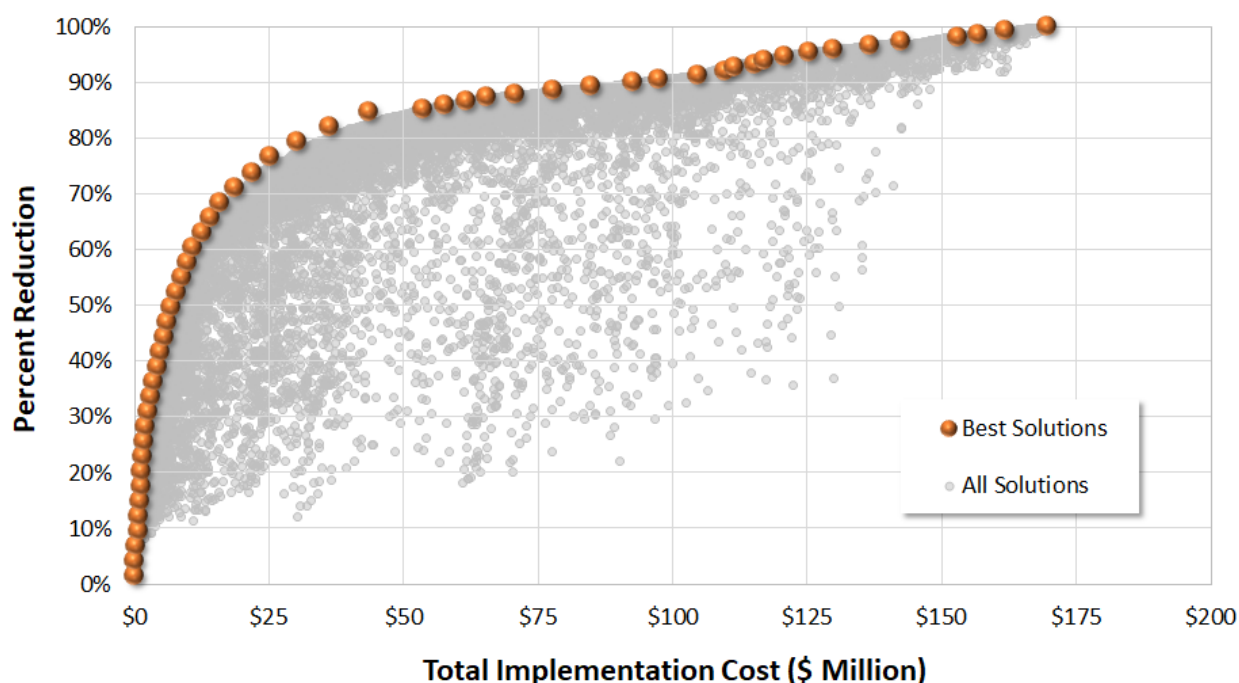


Figure 1-1. Example solutions for a single jurished in the cost-benefit optimization.

By default, each optimization solution in the cloud is independent of every other solution. During an optimization, BMP performance may favor a certain suite of BMPs at lower reduction intervals and a different suite at higher intervals. In practice, it would be cumbersome for long-term implementation planning to implement one suite of BMPs at lower percent reduction, then implement another suite for achieving higher reduction at a later time. Identifying a set of best solutions that is “inclusive” addresses this issue by selecting BMP suites that have the same base BMP configuration. For example, the suite of BMPs that achieves a 10% reduction will all be included in the suite that achieves 50% reduction, the difference being that BMPs in the 50% reduction suite will be larger and/or include additional BMPs. The Inclusive Best Solutions Utility is a post-process that incrementally combs through the Tier 1 cloud to build a solution set where higher-reduction solutions are inclusive of lower

ones. The number of points sampled along this curve is determined by the “management level” variable described in Section 4. For example, the recommended management level of 100 solutions will produce an inclusive Tier 1 curve of 100 best solutions.

Within a jurisdiction, the inclusive set of best solutions represent the most cost-effective BMP implementation options. However, when compared with results from another jurisdiction, it is sometimes more cost-effective on a regional scale to manage more aggressively in another jurisdiction to meet downstream objectives. From this concept, the second tier of the optimization can be applied. With this approach, the varying cost and benefits of Tier 1 curves are balanced across a larger management area to meet downstream objectives most cost-effectively. In the WMMS 2.0 Optimization Framework, this is completed by the Two-tiered Optimization Utility. Figure 1-2 illustrates a conceptual diagram of “N” number of jurisdictions upstream of an assessment point in Tier 1 and how their solutions are incorporated across the Tier 2 search domain to meet the user-defined reduction target.

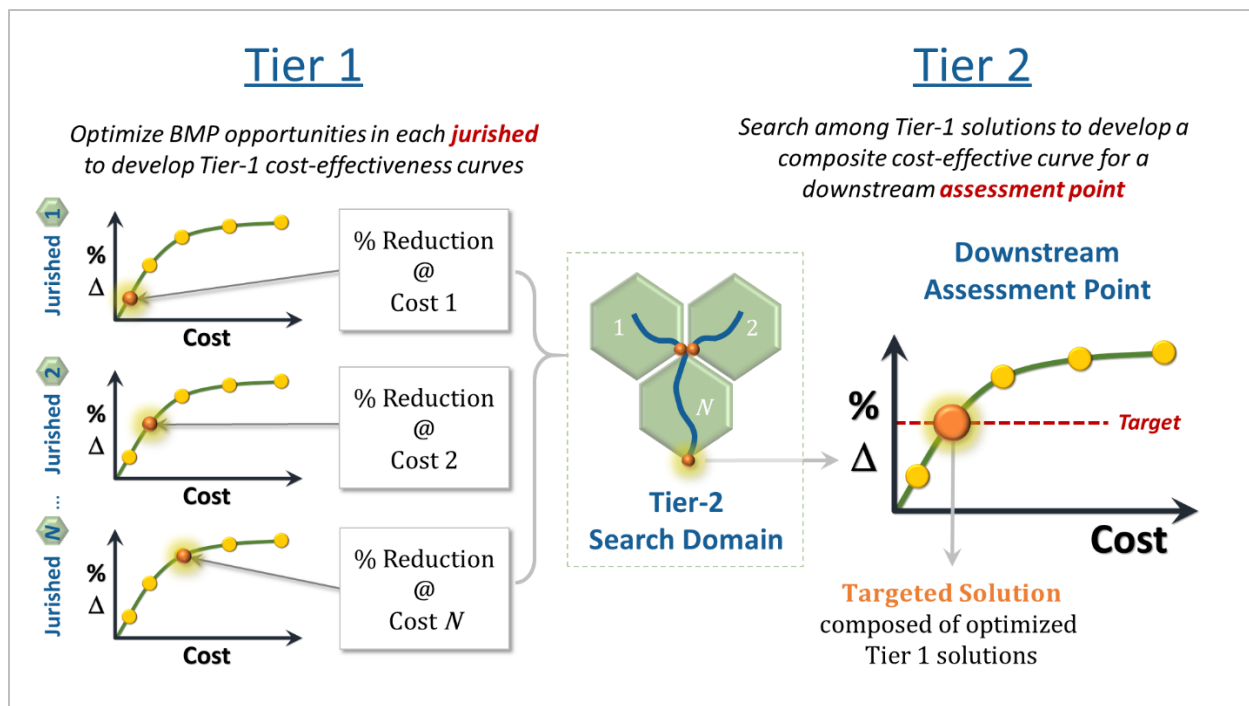


Figure 1-2. Conceptual representation of the two-tier optimization approach.

In simulations where multi-jurisdictional regional (MJR) BMPs are considered, an additional level of optimization is applied. A MJR BMP is a large regional project whose drainage area extends to more than one jurisdiction. The MJR BMP Utility was developed to account for the varying scales at which MJR BMPs performance can be influenced by upstream projects. This utility incorporates each modeled MJR BMP and its upstream components (e.g., drainage area, other BMPs) to balance capture volume, size, and cost across the managed area.

2 TIER 1 OPTIMIZATION

As described in the *BMP Opportunities Utility User Document*, utility output has a very specific organization for the included files and folders when set on a user's computer. The steps below outline the process for preparing the provided files and folders and executing the Tier 1 optimization.

1. The contents of the BMP Opportunities Utility output should be unzipped and placed in newly created folder on the C-drive at C:\MyFiles\WMMS\SUSTAIN\Current. The "Current" folder should always contain the latest output from the utility. When new utility output is generated, the old output should be moved to a different folder with a descriptive name (e.g., "2020_Sept21_Run1"). The new utility output should always be placed in the Current folder to ensure that the most recent version is always being simulated in SUSTAIN.

The LSPC model and 10-year precipitation files available on the WMMS 2.0 Model and Timeseries Repository should be downloaded to develop the HRU unit-area timeseries, which represent the runoff and loading timeseries for one acre of each HRU in each jurished. LSPC should be configured and, if applicable, further calibrated to generate the necessary HRU unit-area timeseries. Output from LSPC will be organized into folders that follow the convention "WSTNUM-XXX", where XXX is a three-digit value from 001 to 906 representing the 906 precipitation zones in LSPC. These folders contain the HRU unit-area timeseries and should be moved to ...\\Current_Input\LSPCData. See Figure 2-1 for an example of folders containing the HRU timeseries and Figure 2-2 for the timeseries contained in each folder.

A potential evapotranspiration text file should also be output from LSPC, named "PEVT.txt", and added to each WTSNUM folder.

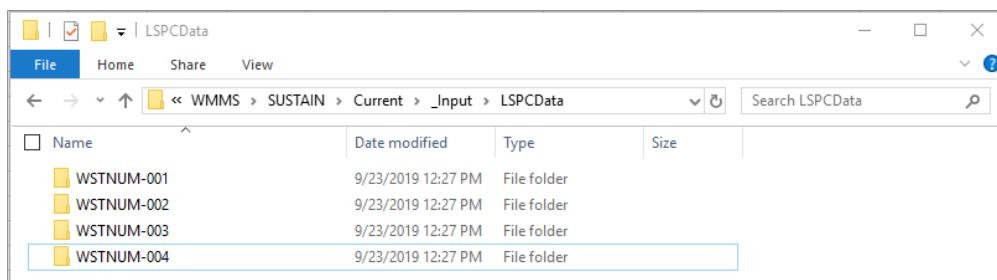


Figure 2-1. Folder structure for output from LSCP organized by the 906 precipitation zones.

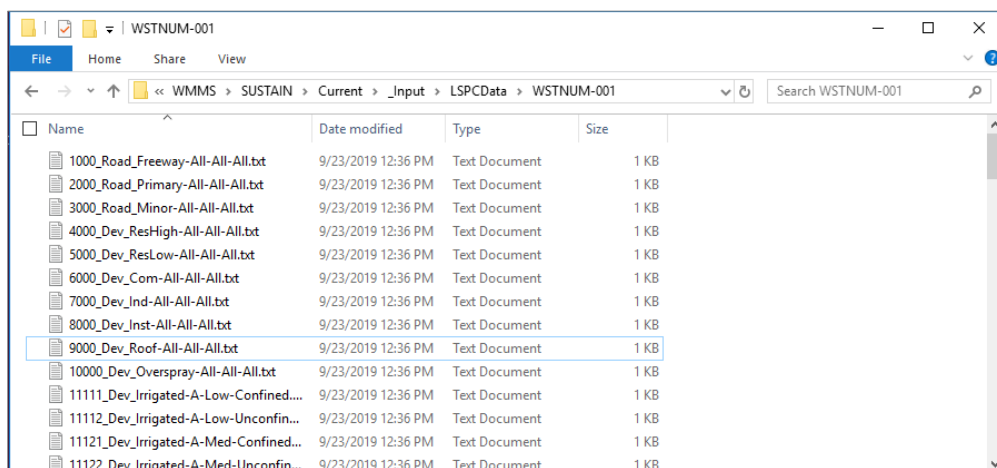


Figure 2-2. LSPC output representing the 96 HRU timeseries.

The SUSTAIN executable (.exe) and dynamic link library (.dll) are also available on the Model and Timeseries Repository and should be downloaded to ...\\Current_Model (Figure 2-3). This is the file path the Optimization Framework expects the model executable to be placed. If future versions of SUSTAIN are introduced, they can be located in this folder and the older model files can be archived.

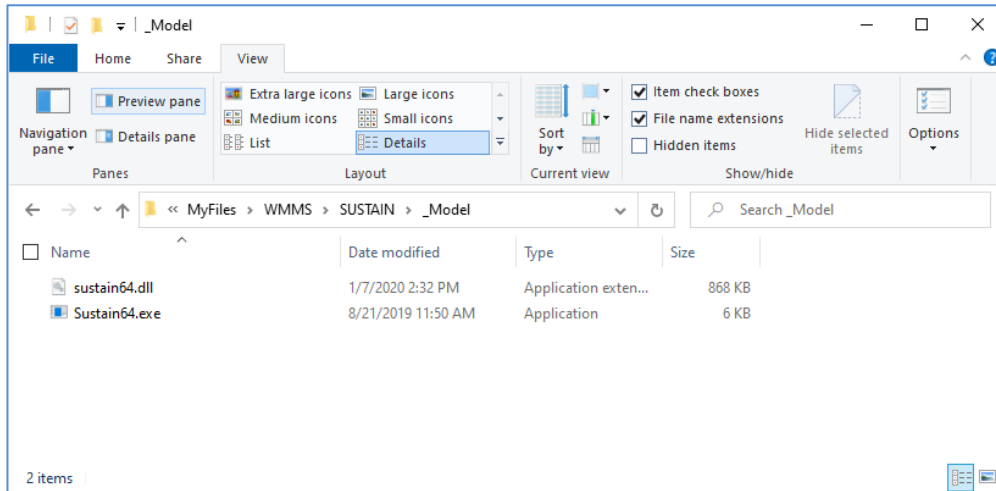


Figure 2-3. Folder directory where SUSTAIN model files should be placed.

A Microsoft C++ Redistributable update, also located on the Model and Timeseries Repository, must be downloaded and installed for SUSTAIN to properly execute. After this has been completed, the user's computer should be restarted to finalize the installation process.

2. Once the above steps have been completed, the SUSTAIN Tier 1 optimization can be executed by double-clicking the included batch (.bat) file in ...\\Current\\1optimized (Figure 2-4).

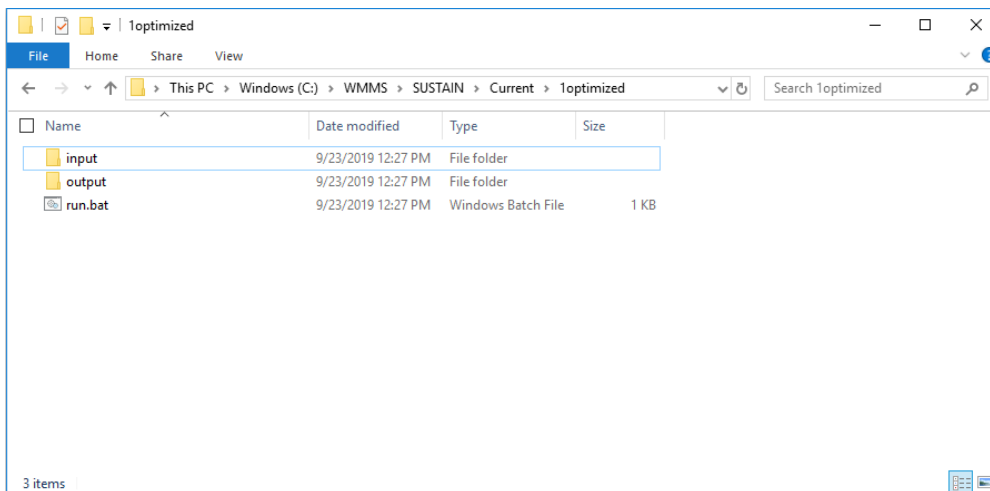


Figure 2-4. Folder directory for batch file and model input and output.

On some computers, executing this batch file will initiate a warning from Windows Defender SmartScreen (Figure 2-5). This warning can be bypassed by clicking on the “More info” button, which gives the option to “Run Anyway”. Once running, the batch file will simulate each SUSTAIN input file in ...\\Current\\loptimized\\input and generate model results to ...\\Current\\loptimized\\output. This process produces the “All Solutions” cloud and “Best Solutions” curve (Figure 1-1) for each modeled jurished and may take several hours or days depending on the number of decision variables (BMPs, jurisheds, simulation period, etc.). Another Windows Warning that may occur on a user’s computer is an “Open File – Security Warning”. This can be by bypassed by clicking the “Run” button on the dialog box.

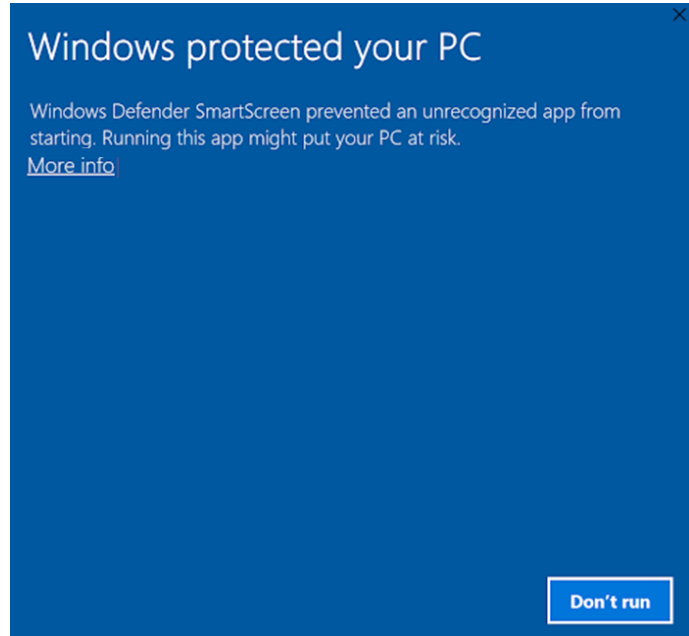


Figure 2-5. Warning from Windows Defender.

3. When the Tier 1 optimization is complete, the output folder should be “zipped” and uploaded to the Inclusive Best Solutions Utility.
4. Also uploaded to the Inclusive Best Solutions Utility is the “data.wmms” file located in the “4viewer” folder (...\\Current\\4viewer). This file contains model specifications from the BMP Opportunities Utility template that must accompany the zipped optimization folder from Step 4.
5. Lastly, enter the maximum percent reduction target for the selected constituent (e.g., total zinc, flow volume). For example, if 100 is entered, 100% reduction will be used as the upper bound in identifying inclusive solutions in each jurished. The recommended value is 100% (i.e., maximum possible reduction).
6. Once these steps have been completed, the utility should be executed by clicking the “Process” button. This will develop a suite of inclusive best solutions from the “All Solutions” cloud for each modeled jurished. This process can take up to 10 or 20 minutes. Once completed, a zipped folder will automatically be downloaded from the utility page.

3 TIER 2 OPTIMIZATION

Once an inclusive set of best solutions has been formed from the Tier 1 cloud, the second tier of optimization can be executed. The steps below outline the process for preparing the input files.

1. Output from the Inclusive Best Solutions Utility should be unzipped to: ...Current\2production\output. Existing folders will already be generated here from the BMP Opportunities Utility. Output from the Inclusive Best Solutions Utility should be placed alongside it.
2. Next, the Tier 2 optimization can be initiated by double-clicking the included batch (.bat) file in ...Current\2production. This process may take several hours or days depending on the number of variables (jurisheds, simulation period).
3. Once the optimization has completed, model output in ...Current\2production\output should be zipped and uploaded to the next utility. As shown in Figure 3-1, the appropriate utility is determined by the presence or absence of MJR BMPs in the model domain.
 - ▼ If MJR BMPs are in any of the considered jurisheds, model output can be uploaded to the MJR BMP Utility. Files that accompany this upload are described in Section 4.
 - ▼ If MJR BMPs are not present in the considered jurisheds, model output can be uploaded to the Two-tiered Optimization Utility. Files that accompany this upload are described in Section 5.

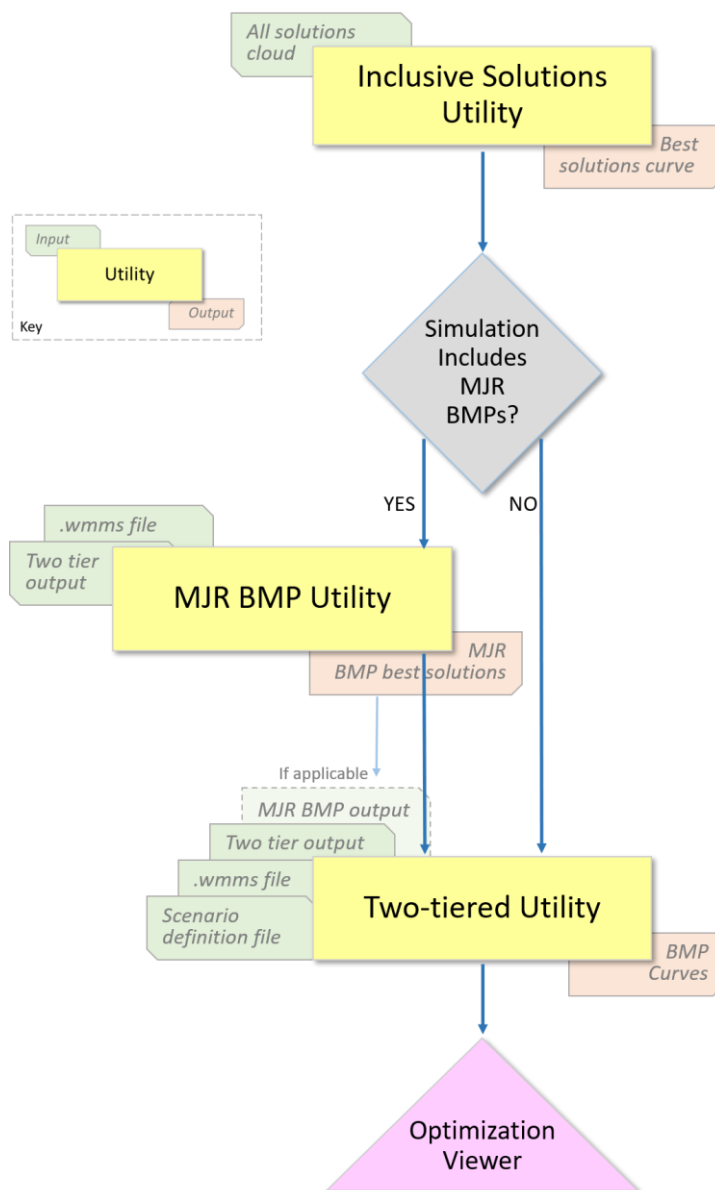


Figure 3-1. Decision tree illustrating the utility path for simulations with and without MJR BMPs.

Figure 3-1 also illustrates the necessary input and output for each of the optimization utilities. Whether MJR BMPs are included or not, the Two-tiered Optimization Utility is the final step before uploading results to the Optimization Viewer.

4 MJR BMP UTILITY

As described in the steps above, when MJR BMPs are present, the MJR BMP Utility is executed after running both the Inclusive Solutions Utility and the Tier 2 optimization. If MJR BMPs are not present, a user should proceed directly to the Two-tiered Optimization Utility described in the next section.

The MJR BMP Utility generates output files for the MJR BMP simulation such that their performance relative to that of the other BMPs in upstream jurisdictions can be evaluated. As needed, the *WMMS 2.0 Phase II Report: BMP Model and Optimization Framework* and the *BMP Opportunities Utility User Document* should be referenced for a full description of how MJR BMPs are determined and simulated. The steps below outline the process for executing the MJR BMP Utility.

1. Once the Tier 2 optimization has completed, model output (in `...\Current\2production\output`) should be zipped and uploaded to the MJR BMP Utility.
2. As with the Inclusive Best Solutions Utility, the `.wmms` file inside the “4viewer” folder (`...\Current\4viewer`) should be uploaded to the space provided on the MJR BMP Utility page.
3. After these two uploads have been made, the utility should be executed by clicking the “Process” button. This will generate the files necessary to run the MJR BMP optimization. This process may take several minutes depending on the number of simulated MJR BMPs. Once completed, a zipped folder will automatically be downloaded from the utility page.
4. Output from the MJR BMP Utility should be unzipped and placed on the user's C-drive at file path: `...\Current\3mjrBmpProduction\output`. Existing folders may already be generated here from the BMP Opportunities Utility. Output from the MJR BMP Utility should be placed alongside it.
5. Next, the MJR BMP run can be initiated by using the included batch (`.bat`) file in `...\Current\3mjrBmpProduction`. Double click the `run.bat` file. This process may take several hours or days depending on the number of decision variables (jurisdictions, model time period). When the simulation is complete, a zipped folder named `Tier2DataPackage.zip` located at `...\Current\3mjrBmpProduction` will be generated.
6. The resulting `Tier2DataPackage.zip` and other necessary files should be uploaded to the Two-tiered Optimization Utility (described in the next section)

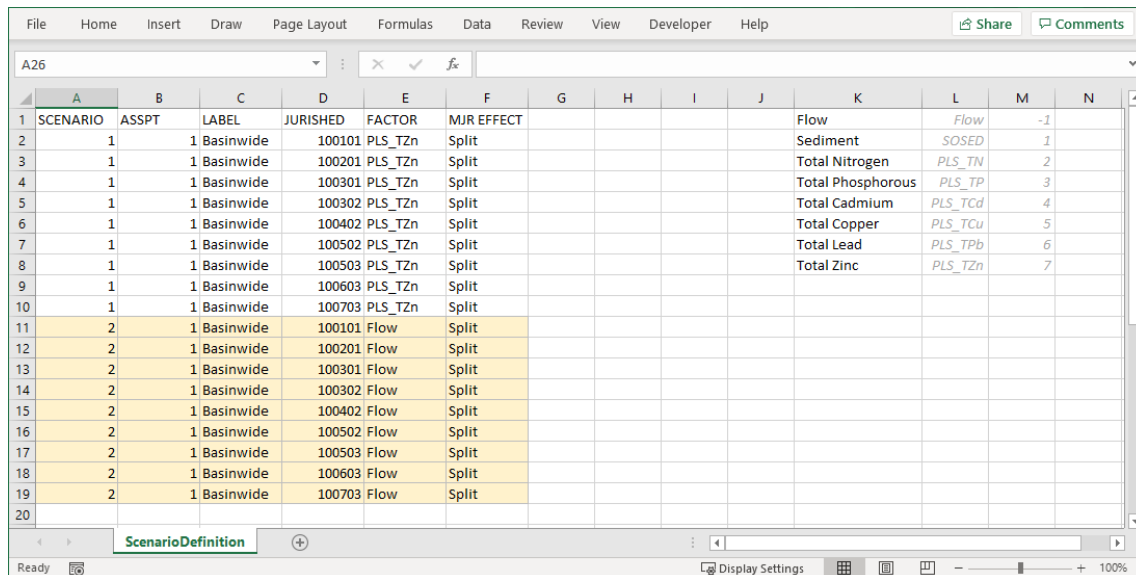
5 TWO-TIERED OPTIMIZATION UTILITY

As shown in Figure 3-1, whether a simulation does or does not include MJR BMPs, output is ultimately uploaded to the Two-tiered Optimization Utility to summarize all modeled results for the Optimization Viewer. The steps below outline the process for running the Two-tiered Optimization Utility.

1. Zipped output should be uploaded to the utility.
 - ▼ If MJR BMPs were considered, output from the MJR BMP simulation at `...\Current\3mjrBmpProduction\Tier2DataPackage.zip` should be uploaded to the Two-tiered Optimization Utility.
 - ▼ If MJR BMPs were not considered, output from the second tier of optimization at `...\Current\2production\output` should be uploaded to the Two-tiered Optimization Utility.
2. Next, a “scenario definition file” should be created and uploaded to the Two-tiered Optimization Utility. A sample file is provided for download on the utility page. This file is in the form of a spreadsheet and defines the decision space and evaluation factors (water quality constituents and/or flow) that are included for assessment in the Optimization Viewer in four columns (Figure 5-1).
 - a. **SCENARIO.** A “SCENARIO” is a grouping of ASSPTs and evaluation factors (FACTOR) for a specific set of jurisdictions. An important detail is that **the jurisdictions associated with a scenario must be hydrologically connected**. This is especially important where MJR BMPs exist within the routing network because mass balance must be preserved. In Figure 5-1, a unique list of jurisdictions is duplicated for two evaluation factors; therefore, two SCENARIOS are defined.
 - b. **ASSPT.** “ASSPT” is an index that corresponds to a grouping of assessed jurisdictions representing a particular location or conceptual association (i.e., “Basin-wide” in the example below denotes that all jurisdictions are part of a basin). In Figure 5-1, ASSPT 1 corresponds to nine jurisdictions. If the user wants to assess a different pollutant (FACTOR), for the same group of jurisdictions, a new SCENARIO should be defined. Figure 5-1 illustrates this for total zinc and flow reduction across nine jurisdictions.
 - c. **LABEL.** A “LABEL” is a short description associated with each SCENARIO and ASSPT combination. It serves as an organizational aid to for the user. In this example, a “Basin-wide” assessment point is being evaluated. For other configurations, LABEL can be used to denote jurisdictions or other jurisdiction associations.
 - d. **JURISHED.** “JURISHED” lists all jurisdictions that are to be considered across an assessment area for a scenario. **The list of jurisdictions need to be hydrologically connected to preserve mass balance.**
 - e. **FACTOR.** “FACTOR” is the constituent being optimized for each SCENARIO. The constituents that can be selected are listed in column L in Figure 5-1. The utility can only process FACTOR values that are among those referenced in column L.
 - f. **MJR EFFECT.** This field specifies how SUSTAIN accounts for the EFFECT of the MJR BMPs. It is used to either share the cost and benefit of MJR BMPs to contributing upstream source areas or to assign both costs and benefit to jurisdiction associated with the physical footprint of the considered MJR. One of the following two values is expected:
 - “Split” means the cost-response-capacity (CRC) of MJRs is shared among upstream jurisdictions based on the amount of runoff originating from each jurisdiction.

- “Stay” means that the entire CRC of MJRs is assigned to the jurished associated with the footprint of the MJR BMP.

If MJR BMPs are not present, the “Split” option should be used for all considered combinations of SCENARIO, ASSPT, JURISHED, and FACTOR.



SCENARIO	ASSPT	LABEL	JURISHED	FACTOR	MJR EFFECT	Flow	Sediment	Total Nitrogen	Total Phosphorous	Total Cadmium	Total Copper	Total Lead	Total Zinc
1	1	1 Basinwide	100101	PLS_TZn	Split	Flow	SOSED	PLS_TN	PLS_TP	PLS_TCd	PLS_TCu	PLS_TPb	PLS_TZn
2	1	1 Basinwide	100201	PLS_TZn	Split								
3	1	1 Basinwide	100301	PLS_TZn	Split								
4	1	1 Basinwide	100302	PLS_TZn	Split								
5	1	1 Basinwide	100402	PLS_TZn	Split								
6	1	1 Basinwide	100502	PLS_TZn	Split								
7	1	1 Basinwide	100503	PLS_TZn	Split								
8	1	1 Basinwide	100603	PLS_TZn	Split								
9	1	1 Basinwide	100703	PLS_TZn	Split								
10	2	1 Basinwide	100101	Flow	Split								
11	2	1 Basinwide	100201	Flow	Split								
12	2	1 Basinwide	100301	Flow	Split								
13	2	1 Basinwide	100302	Flow	Split								
14	2	1 Basinwide	100402	Flow	Split								
15	2	1 Basinwide	100502	Flow	Split								
16	2	1 Basinwide	100503	Flow	Split								
17	2	1 Basinwide	100603	Flow	Split								
18	2	1 Basinwide	100703	Flow	Split								
19	2	1 Basinwide	100703	Flow	Split								
20													

Figure 5-1. Example scenario definition file. The figure shows two scenarios and one “basin-wide” assessment point for Total Zinc and Flow reduction evaluation factors for nine jurisheds.

Once the scenario definition file is completed, it should be uploaded directly to the provided space on the Two-tiered Optimization Utility page.

3. As with the prior utilities, the .wmms file inside the “4viewer” folder (...\\Current\\4viewer) should be uploaded to the space provided on the utility page.
4. If MJR BMPs are not present, toggle the “MJRBMPs slider” to “off”. If MJR BMPs are present, the toggle should be moved to the “on” position. The MJR EFFECT field is still required whether or not MJR BMPs are simulated.
5. Once all uploads have been made to the utility, it should be executed by clicking the “Process” button. This may take several minutes depending on the number of scenarios and decision variables (jurished and simulation area combinations). Once completed, a package of cost-effectiveness curve(s) that considers all BMPs in the model area will be automatically downloaded. This output is a relatively complex three-dimensional matrix of BMP capacities (“C”) with associated responses (“R”) and costs (“C”) for each management scenario. Each solution in the matrix represents a different spatial footprint of BMPs within the jurisheds associated with the assessment point. This “CRC” download can be uploaded to visualize results from the Two-tier Optimization Utility.

6 OPTIMIZATION VIEWER

Output from the Two-tiered Optimization Utility is aggregated by the Optimization Viewer into a summary plot that illustrate the 3 axes of the CRC matrix including the complete range of load or volume reduction represented by all opportunities along the R-axis, the aggregated capacity by BMP type as a cumulative distribution plot on the primary C-axis (capacity), and the associated cost-effectiveness curve as a dark red line on the secondary C-axis (implementation cost). Figure 6-1 provides an example of this plot with six BMP types simulated. Every point, or “slice”, along the plot represents the optimal solution set that achieves the specified target at the assessment point for the lowest cost. Users can slide the vertical slice along the R-axis of the curve to see how cost and capacity change for each solution.

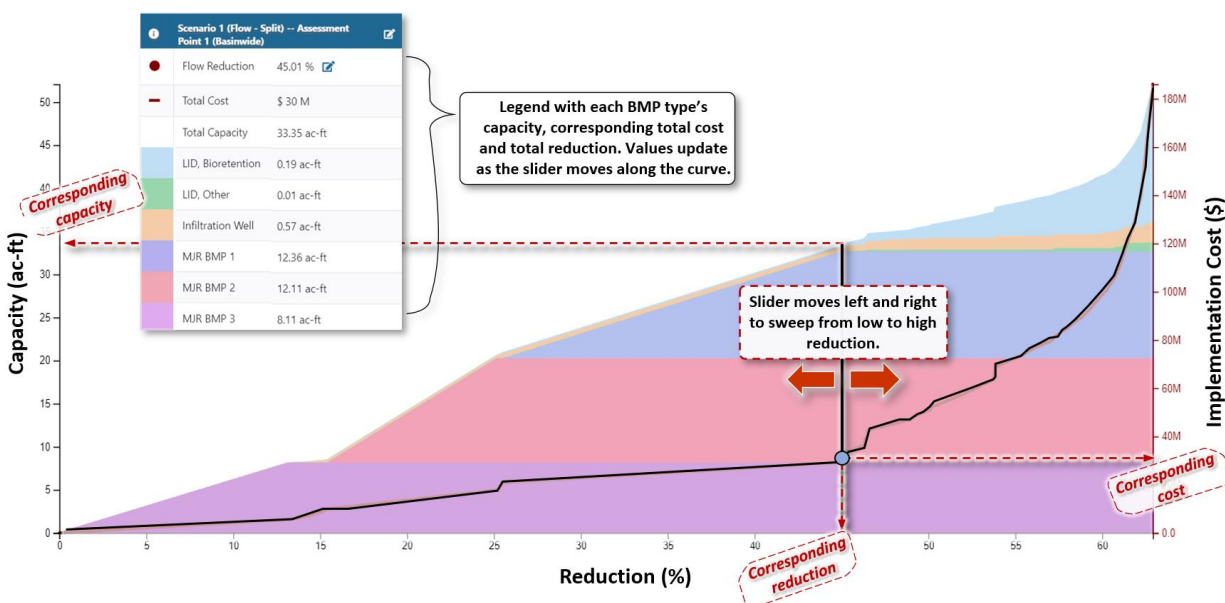


Figure 6-1. Cumulative load reduction summary plot with corresponding BMP types, volumes, and costs.

As described above, the scenario definition file uploaded to the Two-tiered Optimization Utility allows the user to formulate different optimization problems with different conditions and constraints. The list of jurisdictions specified for a SCENARIO must be hydrologically complete so that the mass balance for MJR BMPs can be computed, where applicable. CRC output from the Utility is uploaded to the Optimization Viewer to illustrate BMP performance for the defined scenarios.

Jurisdictional vs Basin-wide Implementation

A common objective for SUSTAIN's two-tiered optimization is to assess the cost and pollutant reduction tradeoffs between jurisdictional and basin-wide BMP implementation strategies. The spatial distribution of implementation cost and capacity is strongly influenced by this formulation. The example scenario definition file from Figure 5-1 has been updated to reflect four scenarios, as shown in Figure 6-2 below. All four scenarios include nine hydrologically connected jurisdictions optimized for total zinc load reduction.

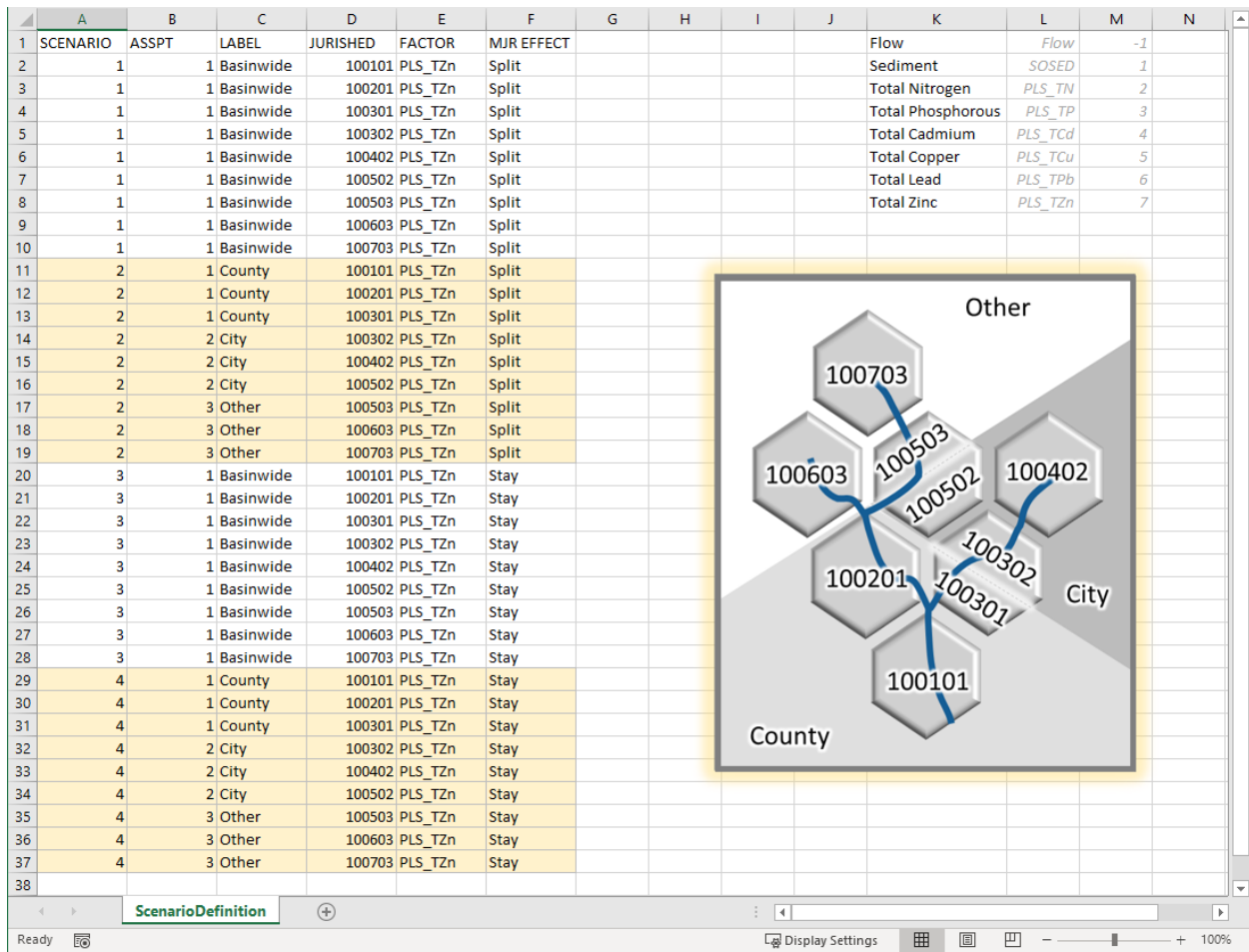


Figure 6-2. Updated example scenario definition file.

Table 6-1 lists a summary table for the scenarios defined in Figure 6-2. Those scenarios can be used to quantify economies of scale associated with an integrated basin-wide strategy vs. each jurisdiction implementing a BMP strategy irrespective of proximal jurisdictions (i.e., “going it alone”). Furthermore, comparison of “Split” vs. “Stay” conditions provides bookends for cost sharing between jurisdictions.

Table 6-1. Scenario definition summary table for Figure 6-2 example

MJR Effect	Assessment Point	
	Basin-wide	Jurisdictional
Split	Scenario 1	Scenario 2
Stays	Scenario 3	Scenario 4

This updated scenario definition file can be uploaded to the Two-tiered Optimization Utility (with all of the original uploads listed in Section 5) to evaluate this alternate perspective. Once completed, a new CRC matrix that considers all BMPs in the model area and the selected jurisdictional areas will be automatically downloaded. Just like before, output can be uploaded to the Optimization Viewer. When uploaded to the Viewer, this CRC will allow the user to visualize the reduction capabilities of the assessed BMPs for all scenarios defined in the scenario definition file. The “Optimization Target,” “Assessment Point,” and “Evaluation Factor” pull-down list are used to toggle between curves from the four scenario outputs (Table 6-2).

Table 6-2. Summary of CRC curves for the Figure 6-2 example

Curve	Scenario	Optimization Target	Assessment Point	Evaluation Factor
1	1	1 - PLS_TZn - Split	1- Basin-wide	Various*
2	2	2 - PLS_TZn - Split	1 - County	Various
3			2 - City	Various
4			3 - Other	Various
5	3	3 - PLS_TZn - Stay	1- Basin-wide	Various
6	4	4 - PLS_TZn - Stay	1 - County	Various
7			2 - City	Various
8			3 - Other	Various

*Evaluation Factors include: Flow, SOSED, PLS_TN, PLS_TP, PLS_TCd, PLS_TCu, PLS_TPb, PLS_TZn. All scenarios were optimized for PLS_TZn in this example; however, *Evaluation Factor* shows secondary benefits to other modeled constituents when PLS_TZn reduction is optimized.

7 UTILITY TROUBLESHOOTING

Various error messages have been integrated into the Utility to indicate potential issues for data formatting or content. As much as possible, errors were written to succinctly and clearly define the issues present so a user could make the appropriate adjustments to their spreadsheet template. The list below provides additional context for errors.

Errors

Message: “Network Error”.

Context: Reload the webpage and try again.

Message: “Error parsing the scenario definition file: No MJR BMP Effect Distribution specified in row [X], No MJR BMP Effect Distribution specified in row [X]”.

Context: The words “Split” or “Stay” is required for every listed jurished in column F of the scenario definition file. Most simulations use the “Split” option.

Message: “Sequence contains no elements”.

Context: Once of the acceptable constituent names in column L of the scenario definition file was not used in column E. For example, to represent total zinc as the target constituent “PLS_TZn” should be used in column E. Reference Figure 6-2 for acceptable constituent names.

Message: “Folders do not exist for all jurisheds specified in scenario definition”.

Context: A jurished listed in column D was not included in prior simulations. Double check that there are no erroneously listed jurisheds.

Message: “Error parsing the scenario definition file: Worksheet 'ScenarioDefinition' does not exist”.

Context: The tab in the scenario definition file should be named “ScenarioDefinition”.